

AKASH V P

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PROFESSIONAL SUMMARY

Associate Software Engineer specializing in Generative AI, AI/ML, and Data Engineering. Experienced in building LLM-powered applications, AI agents, RAG systems, ETL workflows, and large-scale data processing pipelines using Python, LangChain, Gemini, ChromaDB, and MongoDB.

PROFESSIONAL EXPERIENCE

Associate Software Engineer — AI/ML & Data Engineering <i>Bititude Technology</i>	Jul 2025 – Present <i>Kozhikode, Kerala</i>
- Developed LLM-powered applications and AI agents for intelligent search, workflow automation, and information retrieval. - Built production-oriented RAG pipelines using LangChain, Gemini APIs, ChromaDB, embedding models, and Hugging Face Transformers. - Developed scalable data acquisition pipelines that collected, validated, and structured information for 70,000+ organizations. - Designed ETL workflows to extract, transform, validate, and load structured and semi-structured datasets into MongoDB. - Built automated contact intelligence workflows processing 100,000+ records through email pattern generation and enrichment. - Developed Python automation tools for CSV ingestion, database migration, data normalization, and large-scale data updates. - Designed MongoDB schemas and optimized data processing workflows for efficient storage, retrieval, and scalability. - Containerized AI applications using Docker and deployed AI-driven solutions into production environments.	

PUBLICATIONS

A Comparative Study of Machine Learning and Deep Learning Frameworks in Identifying and Detecting Nutrient Deficiencies in Plants using Leaf Image Analysis <i>2025 6th International Conference for Emerging Technology (INCET), IEEE</i>	May 2025
Co-authored with Pushpa B.R and Akash V.V. DOI: 10.1109/INCET64471.2025.11140314	

PROJECTS

AI-Powered Information Retrieval Agent <i>Python, LangChain, ChromaDB, Streamlit, Groq API</i>	Nov 2024
- Developed an LLM-powered retrieval platform supporting document ingestion, web extraction, semantic search, and natural language querying. - Implemented a LangChain-based RAG architecture using ChromaDB and a Streamlit dashboard for interactive information retrieval.	

TECHNICAL SKILLS

Programming : Python, Java, SQL
Generative AI : LLMs, LangChain, RAG, AI Agents, Gemini API, Prompt Engineering, ChromaDB, Embeddings, Semantic Search, NLP
Data Engineering : MongoDB, ETL Pipelines, Data Processing, Data Enrichment, Data Validation, Data Migration, Web Scraping
Machine Learning : TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, CNN, LSTM
Tools : Docker, Git, GitHub, Jupyter Notebook, Hugging Face, Streamlit

CERTIFICATIONS

MCP Course <i>Hugging Face</i>	Jan 2025
Introduction to Natural Language Processing <i>Analytics Vidhya</i>	Mar 2025
Machine Learning using Python <i>Cognitive Class – IBM</i>	Feb 2024

EDUCATION

Amrita School of Arts and Sciences <i>Master of Computer Applications — AI and Data Science</i>	2023 – 2025 <i>CGPA: 8.5</i>
Pazhassi Raja NSS College <i>Bachelor of Science (Physics)</i>	2020 – 2023 <i>CGPA: 6.8</i>

LANGUAGES

Malayalam : Native	English : Professional Proficiency
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